Biliary Atresia, Part Five: The Kasai Operation. Technique, Laparoscopic versus Open Surgery, and Centre Volume.

Imagine a six-week-old boy whose work-up is complete. The stools are pale, the conjugated bilirubin is high, the ultrasound shows a small, irregular gallbladder, and the liver biopsy shows ductular proliferation and bile plugs. He almost certainly has biliary atresia, and he is young enough for the operation to have its best chance. Now a new set of questions appears. His parents have read that the operation can be done through keyhole incisions and ask whether that would be gentler. The local surgeon is skilled but performs perhaps one such operation a year, and the nearest high-volume centre is a four-hour drive away. Does it matter who operates, how the operation is done, and through what incisions? This chapter is about those questions.

The main idea can be stated simply. The Kasai portoenterostomy is a drainage operation, not a reconstruction. It cannot create new bile ducts; it can only open the microscopic channels that the disease has not yet destroyed. Within that limit, the outcome depends on three things: a precise transection of the scar tissue at the porta hepatis, flush with the surface of the liver; avoiding additional harm, such as burning the cut surface, cutting too deep into the liver, or building complicated bowel conduits; and the experience of the whole team, before and after the operation. The incision matters much less than the hands behind it, and so far no approach has beaten the traditional open operation.

A few terms first. The porta hepatis is the hilum of the liver, the place where the bile ducts normally leave it and the portal vein and hepatic artery enter. In biliary atresia the extrahepatic bile ducts are replaced by a fibrous cord that ends in a cone of scar tissue at the porta, sometimes called the fibrous plate. The Kasai portoenterostomy, named after the Japanese surgeon Morio Kasai, who described it in 1959, removes this cord and cone and sews a loop of small bowel onto the cut surface. That loop is called a Roux limb, or Roux-en-Y limb: a segment of jejunum that is divided, brought up to the liver, and joined back to the rest of the bowel further down, so that bile can drain into the intestine. Clearance of jaundice after the operation is usually defined as a total bilirubin below 2 milligrams per decilitre, which is about 34 micromoles per litre, within three months. Native-liver survival means being alive with one's own liver, without a transplant.

The sources for this chapter include the North American multicentre study by Superina and colleagues in the Annals of Surgery in 2011; a single-centre comparison of operative techniques from Juntendo University in Tokyo, published by Abe and colleagues in 2022; the laparoscopic experience from Hannover, published by Ure and colleagues in 2011, and from Hong Kong, published by Wong and colleagues in 2008; the only randomised trial of laparoscopic against open surgery, from Hanoi, published by Son and colleagues in 2023; a 2023 meta-analysis by Okur and colleagues; national cohorts from the United Kingdom, the Nordic countries and Japan; and the guidance documents from the North American and

European paediatric gastroenterology societies, known as NASPGHAN and ESPGHAN, from 2017, and from the American Academy of Pediatrics in 2025.

Part one. What the operation actually does.

To understand why technique matters, it helps to picture what the surgeon is working with. In most infants with biliary atresia the bile ducts outside the liver are a solid cord. At the porta this cord broadens into a cone of fibrous tissue. Inside that cone, however, there are microscopic channels, called residual ductules, that are still connected to the bile ducts inside the liver. They are far too small to see with the naked eye. When the surgeon cuts across the cone exactly at the level of the liver capsule, these ductules are opened, and bile can seep out of them. The Roux limb is then sewn around the edge of the cut surface, so that this seeping bile is collected into the bowel.

This means there is no anastomosis in the usual sense, no joining of one duct to another. The operation depends on capillary-sized channels that the surgeon cannot see and cannot directly protect. Anything that burns, crushes or scars them, or a cut too shallow to reach them, can turn a technically complete operation into a failed one. It also explains why the condition of the liver at the time of surgery, what we might call the substrate, dominates the outcome. The number and size of patent ductules fall as the disease progresses and vary with the anatomical pattern.

The North American study by Superina and colleagues showed this clearly. Among 136 infants in sixteen centres who had a Kasai operation, about 46 percent cleared their jaundice by three months. Clearance was by far the strongest predictor of a good outcome: an infant whose bilirubin fell below 2 milligrams per decilitre by three months had a risk of transplant or death about one twelfth that of an infant who did not, a hazard ratio of 0.08. On the other side of the ledger, the anatomical pattern found at operation mattered most. Infants whose obstruction extended into the ducts at the porta in the patterns that Japanese surgeons call subtypes b, c and d had a risk of transplant or death more than four times higher. Other unfavourable findings were a high level of obstruction, the splenic malformation syndrome, ascites of more than 20 millilitres, a nodular liver, and an age at operation of 75 days or more, each of which raised the risk by roughly two thirds to double.

Part two. The operation, step by step.

The classical open operation, as performed in experienced centres, can be described in six steps, and each step has its own pitfall.

The first step is to confirm the diagnosis and describe the anatomy. If there is a gallbladder with a lumen, the surgeon injects contrast into it to obtain a cholangiogram, which shows whether any part of the duct system is open. A liver biopsy is taken. The surgeon records the level of obstruction, whether there is a cyst in the remnant, and whether there are features of the biliary atresia splenic malformation syndrome, such as several small spleens, a portal vein

that runs in front of the duodenum, an absent inferior vena cava, abnormal position of the organs, or malrotation of the bowel. The pitfall here is an unconfirmed diagnosis: a patent distal duct on the cholangiogram changes the operation, and an infant whose cholestasis has another cause should not have a Kasai at all.

The second step is to mobilise the liver, dividing the ligaments that hold it so it can be lifted and turned out of the wound, which gives a direct view of the porta. The third step is to dissect the fibrous remnant up to the bifurcation of the portal vein, tying off the tiny venous branches that run from the portal vein into the remnant. The fourth step is the heart of the operation: the fibrous cone is divided sharply, flush with the surface of the liver, extending sideways to the points where the right and left hepatic arteries enter the liver. The fifth step is reconstruction: a Roux limb of about 40 centimetres is brought up behind the colon and sewn widely around the edge of the transected plate, with the back row of stitches placed first. The sixth step is to secure haemostasis, usually by simple pressure or with gauze soaked in dilute adrenaline, then to place a drain and return the liver to its position.

The fourth step is where the operation is won or lost. Diathermy, the electrical cautery used to stop bleeding elsewhere, can seal the ductules on the cut surface and should not be used there. Deep stitches placed into the transected surface can occlude them. Dissection into the liver tissue itself creates scar and pockets of bile. And an anastomosis that is too narrow leaves part of the draining surface outside the limb.

Part three. How deep to cut.

For decades surgeons debated whether a deeper and wider dissection, reaching more of the porta, would open more ductules. Some centres adopted an extended portoenterostomy, cutting more widely and more deeply into the hilum. The most informative recent comparison comes from Juntendo University. Abe and colleagues described 55 consecutive infants: 18 operated with the extended technique in an earlier period, and 37 with a modified technique in which the remnant was cut flush with the liver and the sutures were kept shallow and away from the surface that carries the ductules.

The bar chart for this study tells a striking story. Jaundice cleared completely in about 78 percent of infants after the modified operation, compared with 50 percent after the extended one. More importantly, about 70 percent of infants in the modified group were alive with their own liver and free of jaundice at follow-up, compared with only about 22 percent in the extended group. In other words, when the same team moved from a deep, wide dissection to a flush cut, roughly three times as many infants kept a jaundice-free native liver. There was a price: small collections of bile at the porta, called bile lakes, appeared in about one in five infants after the modified operation and in none after the extended one.

This was a single-centre study, not randomised, with small numbers and two different eras, so it cannot prove the point. It fits, though, with other experience. A series of 31 infants operated with an extended technique in Chennai, India, with a mean age of 73 days, reported clearance

in about 45 percent and cholangitis in about 36 percent, which is no better than standard results. Other single-centre reports using careful sharp division of the plate report clearance of 77 to 81 percent. The principle that emerges is that wider and deeper is not reliably better, and that the ductules should be opened and then left alone.

Part four. The Roux limb and its alternatives.

After the porta comes the bowel. The standard Roux limb is about 40 centimetres long, with a range of about 30 to 60 in different centres, brought up behind the colon and joined end to side to the jejunum lower down. Since the 1970s surgeons have tried many modifications to prevent cholangitis, the infection of the bile ducts that is the most common complication after the Kasai. They built antireflux valves into the limb, by telescoping one segment of bowel into another or creating a spur. They made stomas and external conduits, bringing the limb out through the skin to divert bowel content. None of these reduced cholangitis, and the stomas brought new problems: varices at the stoma site, fluid losses, and dense adhesions that made a later transplant harder. They have largely been abandoned. A Chinese randomised trial suggested that an individually shortened limb gives reflux protection similar to the standard length, but this has not changed practice.

Why do mechanical tricks not prevent cholangitis? Because cholangitis after the Kasai is not simply a matter of bowel content flowing back up the limb. It reflects bacteria gaining access to a poorly draining, inflamed and scarred biliary tree. It is very common: in a consortium of nine American children's hospitals, about 57 percent of 223 infants had at least one episode in the first year, and published series report rates of 30 to 60 percent. The practical rule is to keep the reconstruction simple and friendly to a future transplant: one Roux limb of standard length and no stoma.

Not every infant needs a jejunal limb. In a minority, the gallbladder, cystic duct and common bile duct are open down to the duodenum on the cholangiogram, and only the ducts at the porta are blocked. In this situation the surgeon can sew the gallbladder itself onto the porta, an operation called a gallbladder Kasai or portocholecystostomy. Its appeal is that the natural valve of the bile duct, the sphincter of Oddi, is preserved and bowel bacteria have further to travel. In a single-institution series of 97 such operations reported by Hery and colleagues in 2017, about 46 percent of children were free of jaundice with their native liver at five years. The risks are bile leak, kinking of the narrow cystic duct, and obstruction of a small distal duct, so most centres use it only when the anatomy clearly allows. Cystic biliary atresia, in which a cyst forms within the remnant and is often seen on antenatal ultrasound, is another favourable variant: these infants tend to be operated earlier and do better than those with the usual type, and the condition must be distinguished from a choledochal cyst.

Part five. Laparoscopic and robotic surgery.

In the early 2000s surgeons began to perform the Kasai through laparoscopic, or keyhole, incisions. The reasons were appealing. The camera gives a magnified, well-lit view of the

porta, exactly where the critical step takes place. The wounds are small, blood loss is lower and feeding can resume earlier. And because about half of children with biliary atresia will eventually need a liver transplant, fewer adhesions could make that later operation easier.

The last benefit has held up best. A group from Nagoya, Japan, led by Shirota, compared children who had open or laparoscopic Kasai operations. About two thirds of open patients, 31 of 46, and about three quarters of laparoscopic patients, 23 of 30, were alive with a jaundice-free native liver at one year; at two years the figures were 63 percent and 70 percent. These differences were not statistically significant, so the laparoscopic results matched the open ones in this experienced centre. When transplant followed, the laparoscopic patients had fewer adhesions, a shorter recipient hepatectomy and a shorter stay after transplant.

The early Western experience was very different. In Hannover, Germany, Ure and colleagues followed a prospective series of laparoscopic operations and compared them with infants operated open in the preceding years. The study was stopped after an interim analysis of the first twelve laparoscopic patients, because survival with the native liver at six months was lower after laparoscopy, and longer follow-up to twenty-four months confirmed the difference. In Hong Kong, Wong and colleagues reported that six of nine infants operated laparoscopically, two thirds, needed a transplant within one year, compared with about 39 percent after open surgery, and they saw a volvulus and an internal hernia of the Roux limb. In the North American network only five laparoscopic operations were performed, and the authors were cautious. Both the German and Hong Kong groups returned to open surgery. The proposed reasons included a shallower dissection of the plate when working through ports, heat injury from energy devices, effects of the carbon dioxide used to inflate the abdomen, and a long learning curve.

The only randomised trial comes from Hanoi, Vietnam. Son and colleagues randomised 122 infants with the common, type three form of atresia to laparoscopic or open operation between 2009 and 2017, with all operations done by one surgeon using the same principles. The median age at surgery was about 80 days, which is late, as in many health systems. Six months after surgery, about 61 percent of the open group and about 53 percent of the laparoscopic group were free of jaundice. After ten years, survival was about 59 percent after open surgery and about 44 percent after laparoscopic surgery. Neither difference reached statistical significance, with p values of 0.23 and 0.09, but every estimate favoured the open operation, and a gap of almost 15 percentage points at ten years is clinically large. A trial of this size was able to exclude only a very big difference.

Pooled data tell a similar story. The meta-analysis by Okur and colleagues combined 23 studies with 689 laparoscopic and 818 open operations. Laparoscopy reduced blood loss by about 18 millilitres and allowed feeding about three days earlier, but took about 33 minutes longer. There was no difference in clearance of jaundice, cholangitis, overall complications or transplant-free survival at two years. These studies were almost all retrospective, mostly from

high-volume Asian centres, with large differences between them and younger infants in the laparoscopic groups. They show that experts can do the operation laparoscopically without obvious harm, not that laparoscopy improves drainage.

Robotic surgery is the newest variant. A Chinese series of 74 infants reported by Guo and colleagues found jaundice clearance at six months of 75 percent after robotic surgery and 68 percent after open surgery, with native-liver survival of 67 and 63 percent, the latter difference not significant; the robotic operations took about 40 minutes longer. These are early, retrospective results.

Putting it together, open surgery remains the reference standard. Laparoscopy offers small comfort gains in an infant and probably an easier transplant later, but it has not improved survival with the native liver, and the early Western experience was worse. It belongs in high-volume centres that audit their own native-liver survival.

Part six. Centre volume.

If the incision matters little, the team matters a great deal. In the United Kingdom and Ireland in the mid-1990s, McKiernan and colleagues studied all 91 infants operated in two years. In centres doing more than five operations a year, about 61 percent of children were alive with their own liver at five years; in the smaller centres, about 14 percent were. After England and Wales centralised all care in three centres in 1999, the national results became the benchmark: among 443 infants operated between 1999 and 2009, with a median age at surgery of 54 days, 55 percent cleared their jaundice, native-liver survival was 46 percent at five years and 40 percent at ten, and overall survival was about 90 percent.

The Nordic countries tell the same story. Pakarinen and colleagues described 154 infants operated between 2005 and 2016 at a median age of 64 days. Bilirubin normalised in 64 percent, native-liver survival at five years was 53 percent, and overall survival was 88 percent. In this cohort, operation before 65 days and a centre caseload of more than three infants a year independently predicted long-term survival with the native liver. An Indian consensus suggested at least six operations a year per centre. North America has no mandated centralisation, and outcomes vary widely between hospitals: in the North American network, native-liver survival was about 54 percent at one year and 47 percent at two years, and in the nine-hospital Western consortium about 41 percent of infants were alive with their native liver.

Japan is an instructive counterexample. Without formal centralisation, but with a long tradition of specialised paediatric surgeons, the national registry of 3,951 children reports native-liver survival of about 51 percent, 44 percent and 41 percent at ten, twenty and thirty years, with overall survival of about 86 to 89 percent.

Why does volume matter so much? Volume is a team, not only a surgeon. A high-volume centre runs diagnosis and surgery in parallel rather than one after the other. A small number of surgeons perform a standard, flush transection with a simple limb. The hepatology team has a

low threshold to take blood cultures and start intravenous antibiotics for cholangitis. Nutrition is managed actively, with formula rich in medium-chain fats, fat-soluble vitamin supplements and early tube feeding. And transplant is available on site, so a child whose operation fails can be listed promptly, without delay between hospitals. A surgeon doing one Kasai a year cannot develop a feel for the plane at the porta, and a hospital seeing one infant a year cannot sustain the hepatology that follows. Native-liver survival is the product of the operation and of five years of aftercare.

Part seven. What the guidelines say.

No major adult liver society has a guideline on the Kasai operation itself. The 2014 practice guideline on evaluation of the child for liver transplantation, endorsed by the American Association for the Study of Liver Diseases, known as AASLD, supports the Kasai as the first operation, with transplant when it fails or when the liver is already decompensated; on this point all sources agree. The European Association for the Study of the Liver, known as EASL, has no biliary atresia guideline. The 2017 NASPGHAN and ESPGHAN guideline on cholestatic jaundice in infants and the 2025 clinical report from the American Academy of Pediatrics focus on early recognition and urgent referral to an experienced centre, but set no minimum caseload. National policy goes further: England uses three centres for the whole nation, and Nordic data point to a threshold of more than three cases a year. This is the material difference: every national cohort that has measured centre volume has found an effect, yet no society sets a number, leaving the decision to health systems. On open versus laparoscopic surgery the guidelines are silent; open surgery is standard in Europe, and laparoscopy is used in some Asian centres. On the details of the dissection and the limb, there is no guidance at all, only surgical convention.

Part eight. Traps and pearls.

Several errors turn a correctly timed Kasai into a failed one. Using diathermy on the plate seals the ductules; pressure and adrenaline-soaked gauze are the answer. Cutting deeper into the liver adds scar without adding drainage. Adding valves, stomas or conduits does not prevent cholangitis and complicates transplant. A redo Kasai should be considered only in the rare child who had good initial bile flow that stopped abruptly; otherwise the next step is transplant assessment, because repeated surgery at the porta makes the later transplant harder. Choosing keyhole surgery in a low-volume unit trades a smaller scar for a higher risk of losing the native liver. And operating on a decompensated older infant, with ascites and a clotting problem that does not correct with vitamin K, only delays a transplant that is already needed.

Return now to the six-week-old boy. If the local surgeon performs one Kasai a year, transfer to a high-volume centre within days is worth the journey; the delay should be days, not weeks. If the parents ask for a keyhole operation, the honest answer is that open surgery remains the standard, and that laparoscopy is reasonable only in a centre that can show its own native-liver results. If the cholangiogram shows an open distal duct, a gallbladder Kasai is an option. If he

has the splenic malformation syndrome with a preduodenal portal vein, the Kasai still goes ahead, but the anatomy must be recorded and transplant planning starts early. If his bilirubin is still 4 milligrams per decilitre at three months, he should be referred for transplant assessment straight away. If he clears and later develops fever with new jaundice, cholangitis is the first diagnosis and is treated first; a redo operation is rarely the answer. And if he had presented at five months with ascites and an uncorrectable coagulopathy, transplant assessment would come first.

So, tomorrow in clinic, carry these points. The Kasai drains surviving microscopic ductules and cannot create ducts. The plate should be transected flush with the liver capsule, without diathermy on the cut surface; at Juntendo, a flush rather than extended dissection was associated with jaundice-free native-liver survival of about 70 percent instead of 22 percent. One Roux limb of about 40 centimetres is enough; valves, stomas and conduits did not reduce cholangitis, which affects 30 to 60 percent of children. Laparoscopy has not improved native-liver survival: Hannover and Hong Kong abandoned it after worse early results, and in the Hanoi randomised trial ten-year survival was about 44 percent after laparoscopy and 59 percent after open surgery. Pooled data show slightly less blood loss and earlier feeding, at the cost of a longer operation and no gain for the liver. Centre volume matters: about 61 percent against 14 percent native-liver survival at five years in larger and smaller British centres in the 1990s, and a threshold of more than three cases a year in the Nordic countries. And bilirubin below 2 milligrams per decilitre by three months, which cuts the risk of transplant or death to about one twelfth, is the operation's report card.